

Renal Tubular Acidosis, Distal, 3, With or Without Sensorineural Hearing Loss (dRTA-3)

Disease knowledge-base report MONDO:0011268 · OMIM #602722 · Gene: ATP6V0A4 (HGNC:865) · Category: Mendelian, autosomal recessive

Summary

Renal tubular acidosis, distal, type 3, with or without sensorineural hearing loss (**dRTA-3**) is a rare Mendelian, autosomal-recessive kidney tubular disorder caused by **biallelic loss-of-function mutations in ATP6V0A4** on chromosome 7q34. *ATP6V0A4* encodes the **a4 subunit of the membrane-integral V0 domain of the vacuolar-type H⁺-ATPase (V-ATPase)**, a proton pump concentrated on the apical (luminal) membrane of **α-intercalated cells (type A intercalated cells)** in the renal collecting duct. When this pump fails, the distal nephron can no longer secrete protons into the tubular lumen, so the urine cannot be acidified below pH ~5.5 even during systemic acidosis. The result is a lifelong **hyperchloremic, normal-anion-gap metabolic acidosis** with hypokalemia, hypercalciuria, hypocitraturia, nephrocalcinosis, growth failure, and rickets. Because the identical pump also acidifies the endolymph of the inner ear, affected patients may additionally develop **sensorineural hearing loss (SNHL)** — historically thought to be late-onset (hence "with or without SNHL"), but now known to be able to appear in early childhood as well.

The disease is diagnosed biochemically (inappropriately alkaline urine, pH >5.5, during a normal-anion-gap metabolic acidosis, confirmed by ammonium-chloride acid-loading or furosemide/fludrocortisone testing) and molecularly (next-generation sequencing gene panels or whole-exome sequencing targeting *SLC4A1*, *ATP6V1B1*, *ATP6V0A4*, *WDR72*, *FOXI1*, and *CA2*). Treatment is **lifelong oral alkali** (sodium/potassium bicarbonate and citrate); a prolonged-release potassium citrate/potassium bicarbonate formulation (**ADV7103 / Sibnaya**) improves metabolic control, lowers lithogenic risk, and improves bone density, growth and quality of life relative to standard-of-care alkali. Correcting the acidosis does not reverse established hearing loss, so regular audiological follow-up is essential.

dRTA-3 sits within a genetically heterogeneous family of inherited dRTAs. *ATP6V0A4* accounts for roughly 14–24 % of genetically solved cases in recent multicenter cohorts. Its closest genetic sibling, *ATP6V1B1*-associated dRTA (dRTA type 2 with early deafness, OMIM #267300), encodes the B1 subunit of the same pump and typically produces earlier and more frequent hearing loss; nevertheless, hearing status alone does not reliably distinguish the two genes, which is why molecular testing independent of audiometry is recommended. An *Atp6v0a4* knockout mouse faithfully recapitulates the renal biochemistry and the deafness (with an absent endocochlear potential), and also reveals extrarenal a4 expression in bone, nose, eye and skin.

Section-by-Section Report

1. Disease Information

Overview. dRTA-3 is a primary (genetic) form of **distal (type 1) renal tubular acidosis**, defined by the inability of the distal nephron to maximally acidify the urine during systemic metabolic acidosis, owing to dysfunction of the α -intercalated cells of the collecting duct. It presents in infancy/early childhood with failure to thrive, polyuria, metabolic acidosis, hypokalemia and nephrocalcinosis, and may be accompanied by sensorineural hearing loss.

Key identifiers. - **OMIM:** #602722 (phenotype); gene *ATP6V0A4* OMIM 605239 - **MONDO:** MONDO:0011268 - **Gene / HGNC:** *ATP6V0A4*, HGNC:865; NCBI Gene 50617; Ensembl ENSG00000105929; UniProt Q9HBG4; cytoband 7q34 - **Orphanet:** distal renal tubular acidosis, ORPHA:18 (parent term) - **ICD-10:** N25.89 (disorders resulting from impaired renal tubular function) — no dedicated dRTA-3 code; **ICD-11:** GB90.4 (renal tubular acidosis) - **MeSH:** Acidosis, Renal Tubular (D000141)

Synonyms / alternative names. Recessive distal renal tubular acidosis (rdRTA2); distal RTA with or without sensorineural hearing loss; RTADR (renal tubular acidosis, distal, with deafness); *ATP6V0A4*-related dRTA. The gene was originally named **ATP6N1B** and described as causing dRTA "with preserved hearing" (PMID: 10973252).

Source of information. This report is derived from **aggregated disease-level resources** — OMIM, published cohort studies and case series, functional/model-organism studies, and clinical trials — rather than from individual-patient EHR data.

2. Etiology

Primary cause — genetic. dRTA-3 is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function mutations in *ATP6V0A4***. Smith et al. (2000) identified the gene in dRTA kindreds, finding homozygous truncating (nonsense, deletion, splice-site) mutations in 8 of 9 kindreds with (initially) normal audiometry (PMID: 10973252). The gene "encodes an 840 amino acid novel kidney-specific isoform of ATP6N1A, the 116-kD non-catalytic accessory subunit of the proton pump." (Finding F001)

Genetic risk factors. - **Causal variants:** loss-of-function alleles in *ATP6V0A4* (nonsense, frameshift, splice-site, and deleterious missense). See Section 4. - **Modifier genes:** genetic background modifies the *hearing* phenotype. In V-ATPase-deficient mice, a chromosome-13 modifier locus accounts for ~20 % of hearing-threshold variation (Tian et al. 2017, PMID: 28934385); the renal acidification defect, by contrast, is fully penetrant. (Finding F010)

Environmental / demographic risk factors. - **Consanguinity / family history** is the dominant demographic risk factor because the disorder is recessive: 23 of 26 kindreds were consanguineous in Stover et al. 2002 (PMID: 12414817), and 18 of 39 in Vargas-Poussou et al. 2006 (PMID: 16611712). (Finding F012) - **Sex:** no strong sex bias (≈ 52 % female in the European cohort), as expected for autosomal recessive inheritance (PMID: 30773598). - There are **no established acquired/toxic causes of dRTA-3 itself**; environmental exposures instead cause *secondary/acquired* dRTA that must be distinguished from the genetic disease (Section 13).

Protective factors. No specific genetic or environmental protective alleles are established for dRTA-3. The only "modifiers" documented act on the hearing phenotype (above). Early diagnosis and adequate alkali therapy are protective against complications (growth failure, CKD progression, stone disease) but do not modify disease occurrence.

Gene–environment interactions. None specifically documented for dRTA-3. The disease is essentially fully genetically determined; environmental factors (e.g., intercurrent illness, dietary acid load) influence the *severity of decompensation* and the ease of achieving metabolic control rather than causing the disease.

3. Phenotypes

Renal / systemic phenotypes (Findings F006, F005):

Phenotype	Type	HPO term	Onset	Frequency	Notes
Renal tubular acidosis (distal)	Lab abnormality	HP:0008341 (distal RTA)	Neonatal/infantile	~100 %	Defining feature
Hyperchloremic metabolic acidosis, normal anion gap	Lab abnormality	HP:0001947 / HP:0000114	Infantile	~100 %	Persistent
Inability to acidify urine (urine pH >5.5)	Lab abnormality	HP:0500018 (abnormal urine pH)	Infantile	~100 %	Diagnostic hallmark
Hypokalemia	Lab abnormality	HP:0002900	Infantile	~74 %	Can cause weakness/paralysis
Nephrocalcinosis	Imaging/clinical	HP:0000121	Childhood	74–88 %	Medullary
Hypercalciuria	Lab abnormality	HP:0002150	Childhood	Common	Lithogenic
Hypocitraturia	Lab abnormality	HP:0003128	Childhood	Common	Lithogenic
Nephrolithiasis	Clinical	HP:0000787	Childhood/adult	~21 % (higher in SLC4A1)	
Failure to thrive / short stature	Clinical sign	HP:0001508 / HP:0004322	Infantile	~74–76 %	Mean adult height SDS –0.57
Rickets / metabolic bone disease	Clinical/imaging	HP:0002748	Childhood	Common	From acidosis + Ca loss
Polyuria	Symptom	HP:0000103	Infantile	Common	
Vomiting	Symptom	HP:0002013	Infantile	Common	
Sensorineural hearing loss	Clinical sign	HP:0000407		~22 % (ATP6V0A4)	See below

Phenotype	Type	HPO term	Onset	Frequency	Notes
			Variable (early– adult)		
Chronic kidney disease	Lab/clinical	HP:0012622	Progressive	35 % children, 82 % adults	Stage ≥ 2

Watanabe (2018) summarizes: "Common clinical features of dRTA include vomiting, failure to thrive, polyuria, hypercalciuria, hypocitraturia, nephrocalcinosis, nephrolithiasis, growth delay, and rickets." (PMID: 30588151, Finding F006).

Sensorineural hearing loss (the defining "with or without" feature). In *ATP6V0A4* disease, SNHL is **less frequent and usually later-onset** than in *ATP6V1B1* disease. Priyadarshini 2025 reported SNHL in 22.2 % of *ATP6V0A4* vs 61.5 % of *ATP6V1B1* patients (PMID: 40232499, Finding F006); the Indian registry found SNHL "strongly associated with *ATP6V1B1* variants (53.8 %)" (PMID: 42622870). However, SNHL can be **congenital, early, or progressive** — 7 *ATP6V0A4* probands developed severe early SNHL between 2 months and 10 years (PMID: 16611712, Finding F008). The hearing loss is frequently associated with **large vestibular aqueduct syndrome (LVAS/EVA)** on imaging (PMID: 37121229, Finding F009).

Progression / severity. The renal acidification defect is stable and lifelong; CKD is slowly progressive. Hearing loss may be stable or progressive. Symptom severity is variable, driven partly by adequacy of metabolic control and (for hearing) genetic background.

Quality of life. dRTA and its treatment significantly impact QoL across education/work, social/family life, and emotional/physical well-being; treatment burden (frequent dosing, palatability) is a major contributor, and switching to twice-daily ADV7103 markedly improved QoL (mean satisfaction 9/10) (PMID: 35346296; PMID: 33635379, Finding F011).

4. Genetic / Molecular Information

Causal gene. *ATP6V0A4* (a.k.a. ATP6N1B, ATP6N2, RTADR, VPP2), 7q34; gene OMIM 605239; disease OMIM #602722. Encodes the **a4 subunit** — an 840-amino-acid, kidney-enriched isoform of the 116-kD non-catalytic membrane subunit of the V-ATPase (Finding F001, F014).

Pathogenic variants. - **Affected gene / IDs:** *ATP6V0A4*, HGNC:865; UniProt Q9HBG4; Ensembl ENSG00000105929 (Finding F014). - **Variant classification (ACMG/AMP):** the majority of reported disease alleles are **pathogenic / likely pathogenic** truncating variants; deleterious missense variants also occur. - **Variant types: nonsense, frameshift (deletions/insertions), splice-site, and missense.** Smith 2000 found homozygous truncating mutations in 8/9 kindreds (PMID: 10973252). Vargas-Poussou 2006 reported 14 new + 5 recurrent *ATP6V0A4* mutations across 21 families (PMID: 16611712). - **Functional consequence: loss of function** of the apical proton pump (Findings F004, F008). Batlle & Haque (2012): *ATP6V1B1* and *ATP6V0A4* mutations "cause loss of function of the apical H⁺-ATPase and autosomal recessive dRTA" (PMID: 23114896). - **Origin: germline**, biallelic. Not a somatic disorder. - **Allele frequency:** individual pathogenic alleles are rare in population databases (gnomAD); no single common founder allele is established for *ATP6V0A4*, although recurrent/founder alleles occur in consanguineous populations.

Modifier genes. Genetic background modifies the inner-ear phenotype (Chr-13 modifier locus in mice, ~20 % of hearing-threshold variance; Tian 2017 PMID: 28934385) (Finding F010).

Epigenetic information. No disease-specific DNA-methylation or histone-modification signatures are established for dRTA-3. Not applicable / not available.

Chromosomal abnormalities. None characteristic; dRTA-3 is a single-gene disorder, not a copy-number/structural syndrome.

Molecular assembly context. The a4 subunit is part of the **membrane-integral V0 domain**; a functional pump requires reversible assembly of the cytosolic **V1 domain** (ATP-hydrolytic, includes B1/*ATP6V1B1*) with V0 (proton-translocating). Eaton et al. 2024 showed the assembly factor **Dmxi1/Rabconnectin-3A** is required for V1–V0 assembly, and intercalated-cell-specific *Dmxi1* knockout produces high urine pH like B1 knockouts: "the V-ATPase domains, V1 and V0, must assemble to produce a functional holoenzyme" (PMID: 38984989, Finding F014).

5. Environmental Information

- **Environmental factors:** none cause dRTA-3 (a monogenic disease). Environmental/toxic/drug exposures instead produce *acquired* dRTA (differential diagnosis, Section 13).
- **Lifestyle factors:** dietary acid load and fluid status influence decompensation and stone risk but do not cause the disease.
- **Infectious agents:** not applicable to genetic dRTA-3. (Note: infections such as visceral leishmaniasis can produce acquired acidification defects — a mimic, not the disease itself.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function mutation in *ATP6V0A4*** (germline) **results in** loss/dysfunction of the a4 subunit of the V0 domain of the vacuolar H⁺-ATPase.
2. Loss of a4 **leads to** failure of V0 assembly/function at the **apical plasma membrane of renal α-intercalated cells**, so a functional proton pump is not delivered to the luminal membrane (inferred to include impaired trafficking/assembly, by analogy to the Dmx11 assembly requirement).
3. Loss of apical H⁺-ATPase activity **results in** inability to secrete protons (H⁺) into the collecting-duct lumen; the basolateral anion exchanger AE1 (SLC4A1) and cytosolic carbonic anhydrase II remain intact but are functionally uncoupled from proton export.
4. Impaired distal H⁺ secretion **leads to** failure to lower urinary pH below ~5.5 during systemic acidosis and reduced net acid excretion (impaired titratable acid + ammonium excretion).
5. Retained acid **results in** systemic **hyperchloremic, normal-anion-gap metabolic acidosis**.
6. Metabolic acidosis and impaired distal transport **lead to** secondary consequences: renal potassium wasting → **hypokalemia**; buffering of acid by bone → calcium and citrate mobilization → **hypercalciuria + hypocitraturia**; low urinary citrate and high urinary calcium → **nephrocalcinosis / nephrolithiasis**; chronic acid load in growing children → impaired bone mineralization → **rickets, growth failure**; chronic nephrocalcinosis/tubulointerstitial injury → **progressive CKD**.
7. **Branch (inner ear):** the same a4-containing V-ATPase acidifies **endolymph in the cochlea/endolymphatic sac**; its loss **results in** disrupted endolymph ion/pH homeostasis and **loss of the endocochlear potential, leading to** sensorineural hearing loss (often associated with enlarged vestibular aqueduct). Onset is variable (early to adult) and modified by genetic background.

Molecular pathways / biochemistry. The core defect is in **acid-base transport**, not a canonical signaling cascade. α-intercalated cells "secrete protons into the tubular lumen through H⁺-ATPases functionally coupled to the basolateral anion exchanger 1 (AE1)"; substrate H⁺ and HCO₃⁻ are generated by cytosolic carbonic anhydrase II ([PMID: 23114896](#), Finding F004). "Dysfunction of intercalated cells in the collecting tubules accounts for all the known genetic causes of dRTA."

Cellular processes. Failure of regulated proton secretion / intracellular and luminal pH regulation; secondary tubulointerstitial injury/nephrocalcinosis. In the inner ear, additional pH-dependent processes are implicated (e.g., autophagic dysregulation of hair cells in a V-ATPase-deficient zebrafish model, [PMID: 38277730](#)).

Protein dysfunction. Loss of function of the $\alpha 4$ subunit; the multisubunit holoenzyme cannot translocate protons across the apical membrane (Finding F014).

Metabolic changes. Systemic acidosis; secondary bone mineral mobilization; hypocitraturia reflecting proximal citrate reabsorption driven by intracellular acidosis.

Biochemical abnormality (defining). An **ion-transporter/proton-pump defect** — the paradigmatic "biochemical abnormality" category for this disease.

Immune involvement. None in the primary genetic disease (immune-mediated α -intercalated-cell injury defines an *acquired* mimic — Sjögren's disease, Section 13).

Ontology suggestions. - **GO (biological process):** proton transmembrane transport (GO:1902600); regulation of intracellular pH (GO:0051453). - **GO (cellular component):** proton-transporting V-type ATPase, V0 domain (GO:0033179); vacuolar proton-transporting V-type ATPase complex (GO:0016471); apical plasma membrane (GO:0016324). - **CL (cell types):** renal alpha-intercalated cell (CL:1000715); kidney collecting duct intercalated cell (CL:1001432); cochlear hair cell. - **UBERON:** kidney collecting duct (UBERON:0001232); cochlea (UBERON:0001844); endolymphatic sac (UBERON:0002400); bone (UBERON:0002481). - **CHEBI:** proton/hydron (CHEBI:15378); bicarbonate (CHEBI:17544); citrate (CHEBI:16947); potassium (CHEBI:29103).

7. Anatomical Structures Affected

Organ level. - **Primary:** kidney (collecting duct / distal nephron) — UBERON:0001232, UBERON:0002113. - **Secondary/other primary target:** inner ear (cochlea, endolymphatic sac) — UBERON:0001844, UBERON:0002400. - **Secondary involvement:** bone/skeleton (rickets, osteopenia; UBERON:0002481); urinary tract (stones); growth axis (short stature). - **Body systems:** renal/urinary; auditory/nervous (sensory); musculoskeletal; endocrine-metabolic (acid–base).

Tissue and cell level. - **Tissue:** renal collecting-duct epithelium; cochlear sensory/supporting epithelium. - **Cells:** renal α -intercalated (type A intercalated) cells — CL:1000715 / CL:1001432; cochlear hair cells and endolymph-regulating epithelial cells. The *Atp6v0a4* β -galactosidase reporter also revealed developmental expression in **bone, nose, eye and skin** (PMID: 22872862, Finding F003).

Subcellular level. Apical plasma membrane (GO:0016324); V-ATPase V0 domain (GO:0033179); the $\alpha 4$ subunit "localizes almost exclusively to the apical surface of α -intercalated cells" (PMID: 10973252, Finding F001).

Localization / lateralization. Renal involvement is **bilateral** (both kidneys); nephrocalcinosis is bilateral medullary. Hearing loss is typically **bilateral**, often with bilateral enlarged vestibular aqueducts.

8. Temporal Development

- **Onset:** typically **congenital/infantile** — median presentation age **0.5 years** (range 0–54) in the European cohort (PMID: 30773598, Finding F012). Children with *ATP6V0A4* (and *ATP6V1B1*) variants present earlier than those with *SLC4A1/WDR72* (PMID: 42622870).
- **Onset pattern:** chronic, often with acute metabolic decompensation during intercurrent illness in infancy.
- **Progression:** the acidification defect is **stable/lifelong**; complications (nephrocalcinosis, CKD, short stature) are **slowly progressive**. CKD stage ≥ 2 rises from 35 % in children to 82 % in adults (PMID: 30773598, Finding F005).
- **Hearing loss course:** variable — congenital, early, or later-onset/progressive; regular audiological follow-up is essential (PMID: 37121229, Finding F009).
- **Remission:** none spontaneously; treatment controls (but does not cure) the biochemistry. Critical window: early diagnosis and alkali therapy protect growth and renal function.

9. Inheritance and Population

- **Inheritance:** autosomal recessive (PMID: 10973252, Finding F001, F012).
- **Penetrance:** complete for the renal acidification defect; **variable expressivity/age of onset** for SNHL.

- **Consanguinity:** major contributor — 23/26 kindreds (PMID: 12414817), 18/39 (PMID: 16611712) (Finding F012).
 - **Founder effects:** recurrent alleles occur in specific/consanguineous populations; no single global founder allele established for *ATP6V0A4*.
 - **Carrier frequency / precise prevalence:** not firmly established; dRTA overall is a rare disease (Orphanet ORPHA:18).
 - **Gene contribution:** *ATP6V0A4* accounts for ~14.2 % of solved dRTA in the Indian registry (PMID: 42622870) and ~22 % in another cohort; the overall genetic-diagnosis yield in dRTA is ~83 % (PMID: 30773598).
 - **Sex ratio:** approximately equal (~52 % female), consistent with AR inheritance.
 - **Geographic distribution:** worldwide; higher observed frequency in populations with high consanguinity. Genetic variant spectrum differs between Western and Asian cohorts (*SLC4A1* dominant in India).
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10. Diagnostics

Biochemical (core criteria). - **Hallmark:** inappropriately alkaline urine (pH >5.5) in the presence of a **normal-anion-gap (hyperchloremic) metabolic acidosis** (PMID: 19158053, Finding F007). - **Confirmatory provocation tests: ammonium chloride (NH₄Cl) acid-loading test** — dRTA patients keep urine pH >5.5 during induced acidosis (PMID: 21464016); **furosemide (± fludrocortisone) test** — normal controls lower urine pH <5.5, dRTA patients fail (sensitivity ~1.0, specificity ~0.89 vs NH₄Cl; PMID: 10862634). - Supporting labs: hypokalemia, hyperchloremia, low serum bicarbonate, hypercalciuria, hypocitraturia.

Imaging. Renal ultrasound/CT for **nephrocalcinosis** and stones; temporal-bone imaging (CT/MRI) for **enlarged vestibular aqueduct** in patients with hearing loss (PMID: 37121229).

Audiology. Pure-tone audiometry and auditory brainstem response (ABR); the mouse model shows elevated ABR thresholds and absent endocochlear potential (PMID: 22872862).

Genetic testing. NGS gene panels or whole-exome sequencing targeting *SLC4A1*, *ATP6V1B1*, *ATP6V0A4*, *WDR72*, *FOXI1*, *CA2* confirm the molecular subtype; single-gene *ATP6V0A4* testing where phenotype is suggestive. Mutation yield ~83 % (PMID: 30773598, Finding F007). Genetic testing is recommended **independent of hearing status** because hearing does not distinguish *ATP6V0A4* from *ATP6V1B1* (PMID: 16611712, Finding F008).

Clinical criteria / differential diagnosis. No formal society scoring system; diagnosis rests on the biochemical triad plus genetics. Differential includes acquired dRTA and other RTA subtypes (Section 13).

Screening. Cascade genetic testing of relatives once the familial variant is known; carrier testing for reproductive counseling.

11. Outcome / Prognosis

- **Survival:** generally **normal life expectancy** with treatment; dRTA-3 is not primarily life-limiting. (The knockout mouse dies without alkalinization, underscoring the importance of therapy.)
- **Morbidity:** substantial if untreated/undertreated — **CKD stage ≥ 2 in 35 % of children and 82 % of adults**; nephrocalcinosis 74–88 %; mean adult height SDS -0.57 ; stunting up to ~ 75 % in some cohorts ([PMID: 30773598](#); [PMID: 42622870](#), Findings F005, F006).
- **Hearing:** SNHL is **irreversible** once established; hence audiological monitoring and early hearing intervention.
- **Prognostic factors: adequacy of metabolic control is the key modifiable prognostic factor** — adequate control (normal bicarbonate + normocalciuria) was achieved in 51 %, more often in higher-GDP countries (67 % vs 23 %), and was associated with greater height and eGFR ([PMID: 30773598](#), Finding F005). Genotype influences hearing and stone risk.
- **Recovery:** the renal biochemistry is fully correctable with alkali; complications are largely preventable with early, sustained control.

12. Treatment

Pharmacotherapy — **mainstay: lifelong oral alkali** (NCIT: alkalinizing agent). - **Sodium bicarbonate / potassium bicarbonate / potassium citrate / sodium citrate**. Median prescribed dose **1.9 (IQR 1.2–3.3) mEq/kg/day** ([PMID: 30773598](#), Finding F005). Potassium-based alkali is preferred to correct concomitant hypokalemia; citrate additionally reduces stone risk. - **Potassium supplementation** as needed for hypokalemia.

Prolonged-release combination (ADV7103 / Sibnaya) — potassium citrate + potassium bicarbonate). EMA-authorized for dRTA. In an open-label non-inferiority trial (n=37), switching from standard of care raised the response rate from **43 % to 90 %** and reduced urine

Ca/citrate below the lithogenic threshold in 56 % of prior non-responders (PMID: 32712761, Finding F011). Over 24 months, bicarbonate/potassium were normal in 69–86 %/83–93 % of patients, adherence ≥ 75 % in 79 %, and QoL improved by ~89 % with only mild GI adverse events in 17 % (PMID: 33635379). Two years of ADV7103 "improved growth and increased spine BMD" (spine BMD z-score $p=0.024$) (PMID: 36529656).

Supportive / rehabilitative. - Hearing rehabilitation: hearing aids; cochlear implantation for severe/profound SNHL. Speech/language support in children. - **Nutrition and growth monitoring;** stone-prevention measures (hydration, citrate).

Advanced/experimental therapeutics. No gene, cell, or RNA-based therapy is approved or in trials for dRTA-3; management is entirely metabolic/supportive. Pharmacogenomics: not applicable.

Treatment strategy. Early diagnosis → titrate alkali to normalize plasma bicarbonate and urinary calcium/citrate → monitor growth, renal function, nephrocalcinosis, and hearing → multidisciplinary follow-up (nephrology, audiology, endocrinology/growth). NCIT suggestions: Sodium Bicarbonate (NCIT:C287), Potassium Citrate (NCIT:C47679), Potassium Bicarbonate, Cochlear Implantation.

13. Prevention

- **Primary prevention:** not possible for a monogenic disease; **genetic counseling** for at-risk (especially consanguineous) families, with carrier/cascade testing and options including prenatal or preimplantation genetic diagnosis once the familial variant is known.
- **Secondary prevention:** early biochemical diagnosis and prompt alkali therapy to prevent nephrocalcinosis, CKD progression, growth failure and bone disease.
- **Tertiary prevention:** sustained metabolic control (ADV7103/alkali) to prevent stones, preserve renal function and bone density; early hearing intervention.
- **Immunization / public-health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** mouse *Atp6v0a4* (NCBI Gene 140470); the gene is evolutionarily conserved across vertebrates (zebrafish V-ATPase orthologs used as models).

- **Model relevance:** the *Atp6v0a4* knockout mouse is the principal animal model (below). No well-characterized spontaneous companion-animal or wildlife dRTA-3 is established in OMIA for *ATP6V0A4*.
 - **Comparative biology:** the α -intercalated-cell proton-secretion mechanism and V-ATPase subunit composition are conserved between human and mouse; the a4/B1 subunits serve the same kidney and inner-ear functions across species (the d2 subunit co-localizes with a4 in intercalated cells and with a3 in osteoclasts — [PMID: 15800125](#)).
 - **Transmission / zoonosis:** not applicable (non-infectious genetic disease).
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15. Model Organisms

Mouse — *Atp6v0a4*^{-/-} knockout (mammalian; principal model). Recapitulates the human disease: "Atp6v0a4(-/-) mice demonstrated severe metabolic acidosis, hypokalemia, and early nephrocalcinosis," and were "severely hearing-impaired, as shown by elevated auditory brainstem response thresholds and absent endocochlear potential" ([PMID: 22872862](#), Finding F003). Mice die rapidly unless alkalinized; heterozygotes are normal until acid-challenged. A β -galactosidase reporter revealed developmental a4 expression in bone, nose, eye and skin in addition to kidney and inner ear. - **Phenotype recapitulation:** excellent for renal biochemistry and deafness/endocochlear-potential loss. - **Limitations:** high early lethality without alkali; extrarenal expression sites (bone/eye/nose/skin) not obviously symptomatic in humans; hearing phenotype is strongly strain/background dependent.

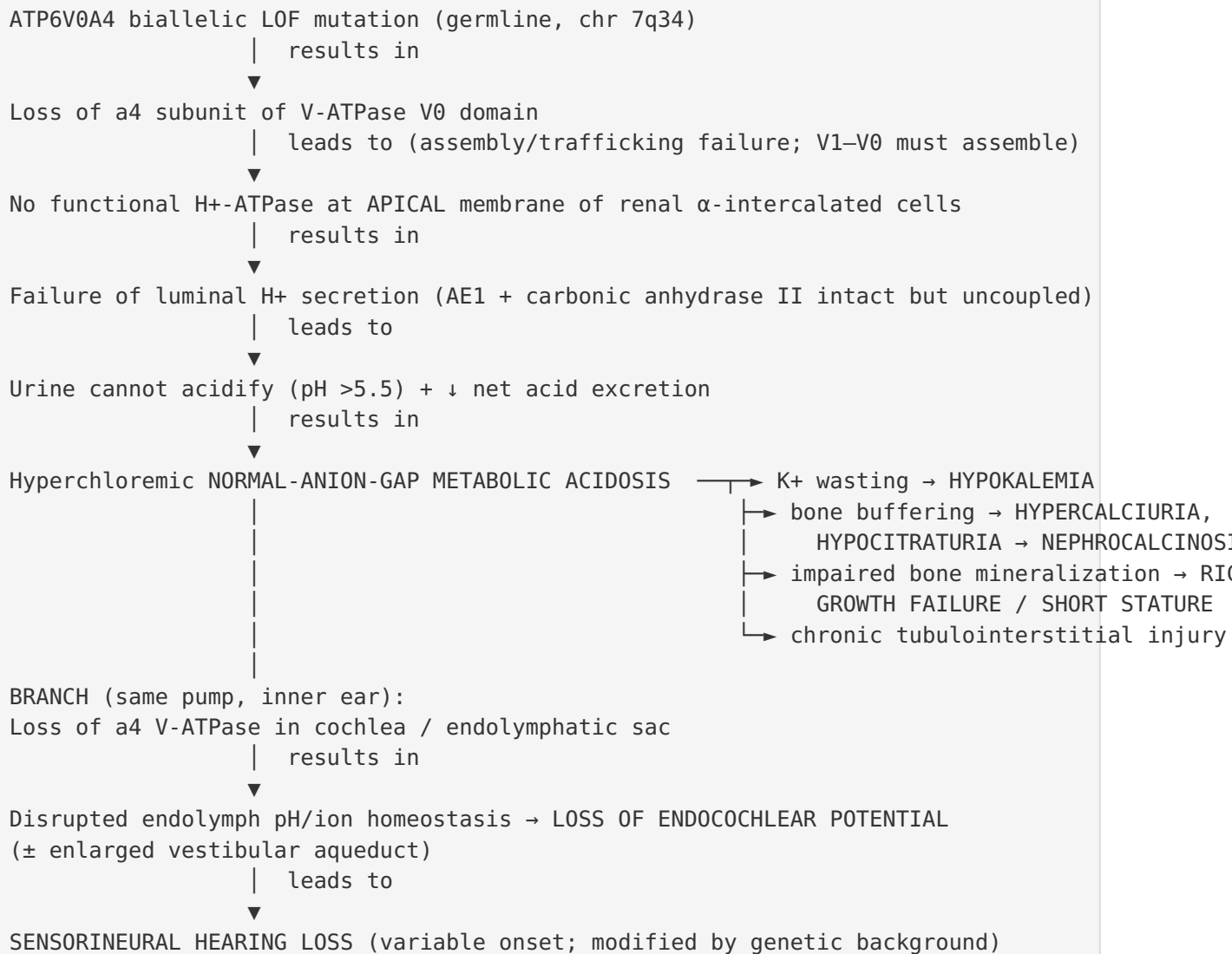
Mouse — *Atp6v1b1* models (sibling gene). MRL-background *Atp6v1b1* mutants show profound hearing loss with enlarged vestibular aqueducts, whereas B6-background knockouts hear normally; a Chr-13 modifier locus explains ~20 % of hearing-threshold variance — a model for the hearing-loss variability of dRTA ([PMID: 28934385](#), Finding F010).

Zebrafish — *atp6v1ba*-deficient. Links V-ATPase-dependent pH imbalance to autophagic dysregulation of inner-ear hair cells ([PMID: 38277730](#)) — mechanistic model of the deafness component.

Mouse — intercalated-cell-specific *Dmxx1* knockout. Demonstrates that V1–V0 assembly (Rabconnectin-3A/Dmxx1-dependent) is required for intercalated-cell V-ATPase function; KO mice have high urine pH like B1 knockouts ([PMID: 38984989](#), Finding F014).

Resources: MGI (mouse), ZFIN (zebrafish), IMPC/IMSR for mutant strains.

Mechanistic Model (synthesis)



Upstream vs downstream: the mutation and pump loss are upstream; acidosis is the pivotal intermediate node; hypokalemia, stone disease, bone disease, CKD and hearing loss are downstream branches. **Alkali therapy acts at the acidosis node** — it corrects the renal/skeletal branches but cannot restore inner-ear proton pumping, explaining why hearing loss is not reversed.

Evidence Base

PMID	Title (abbrev.)	Role in this report
10973252	ATP6N1B mutations cause recessive dRTA	Identifies <i>ATP6V0A4</i> as causal gene; apical α -intercalated-cell localization (F001)
12414817	Novel ATP6V1B1/ATP6V0A4 mutations, hearing loss	Later-onset SNHL in <i>ATP6V0A4</i> ; inner-ear expression; consanguinity (F002, F012)
22872862	<i>Atp6v0a4</i> knockout mouse	Model recapitulates renal + hearing phenotype; absent endocochlear potential; extrarenal expression (F003)
23114896	Genetic causes/mechanisms of dRTA	α -intercalated-cell mechanism; AE1/CAII coupling; LOF (F004)
30773598	Treatment & long-term outcome (European cohort, n=340)	Alkali dose, metabolic control, CKD, height; 83 % mutation yield (F005, F007, F012)
30588151	Improving outcomes in dRTA (review)	Core clinical phenotype list (F006)
40232499	Etiology & outcomes of primary RTA	Genotype-specific SNHL frequency (22.2 % ATP6V0A4) (F006)
16611712	Early SNHL with ATP6V0A4	ATP6V0A4 can cause early deafness; test independent of hearing (F008)
37121229	Hearing loss & gene mutations in dRTA	LVAS association; congenital + progressive hearing loss (F009)
29242249	FOXI1 acidosis + deafness	FOXI1 upstream regulator; genetic heterogeneity (F010)
28934385	ATP6V1B1 MRL mice, EVA	Genetic-background modifier of hearing phenotype (F010)
32712761	ADV7103 comparative trial	Response 43→90 %; lithogenic-risk reduction (F011)
33635379	ADV7103 24-month study	Sustained control, adherence, QoL (F011)
36529656	ADV7103 BMD & growth	Spine BMD z-score $\uparrow$ (p=0.024), growth improved (F011)
19158053		

PMID	Title (abbrev.)	Role in this report
	dRTA due to primary hyperparathyroidism	Diagnostic criterion; reversible secondary cause (F007, F013)
21464016	Primary dRTA case report	NH ₄ Cl acid-loading confirmation (F007)
10862634	RTA in osteopenia/osteoporosis	Furosemide test performance vs NH ₄ Cl (F007)
42694869	Sjögren's dRTA immuno-tubular model	Acquired mimic targeting α -intercalated cells (F013)
42622870	Indian dRTA registry	ATP6V0A4 = 14.2 % of solved cases; SNHL genotype pattern (F006, F012)
38984989	Dmxl1 & V-ATPase assembly	V1–V0 assembly requirement; molecular context (F014)
39150521	WDR72-dRTA	Differential (amelogenesis imperfecta) (F013)
19232111	Osteopetrosis	CA2 mixed RTA differential (F013)
15800125	V-ATPase d2 subunit	Subunit co-localization (α 4 in kidney, α 3 in osteoclasts)
38277730	Zebrafish V-ATPase & hair-cell autophagy	Inner-ear mechanistic model

Limitations and Knowledge Gaps

- 1. Epidemiology is imprecise.** No robust population prevalence, incidence, or *ATP6V0A4* carrier frequency exists; cohorts are consanguinity-enriched and geographically skewed, so genotype proportions (14–24 %) are cohort-dependent.
- 2. Hearing-loss determinants are incompletely understood.** Why some *ATP6V0A4* patients develop early SNHL and others none is unexplained; modifier evidence is largely from mouse strains, and the human genetic modifiers are unmapped.
- 3. No dRTA-3-specific natural-history registry** with long-term renal, auditory and skeletal endpoints stratified by genotype and treatment adequacy.
- 4. No omics data specific to dRTA-3** (transcriptomic/proteomic/metabolomic signatures) were identified; molecular-profiling sections are largely not applicable.

5. **No causal/curative therapy.** Gene- and cell-based approaches are absent; alkali corrects biochemistry but not hearing loss.
 6. **Genotype–phenotype correlation** for individual *ATP6V0A4* variants (e.g., missense vs truncating) is not well resolved.
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Proposed Follow-up Experiments / Actions

1. **Establish an *ATP6V0A4*-specific longitudinal registry** capturing age-stratified audiometry, temporal-bone imaging (EVA), renal function, growth and treatment adequacy to define natural history and genotype–phenotype correlations.
 2. **Prospective audiological surveillance protocol** from diagnosis to detect and time SNHL onset in *ATP6V0A4* patients, testing whether early metabolic control alters hearing trajectory.
 3. **Functional variant characterization** (in vitro assembly/trafficking/proton-pump assays; AlphaFold-guided structural interpretation) to reclassify VUS and correlate residual function with phenotype severity.
 4. **Map human modifiers of the hearing phenotype** by GWAS/whole-genome analysis within dRTA cohorts, building on the murine Chr-13 modifier finding.
 5. **Real-world comparative-effectiveness study of ADV7103 vs conventional alkali** on hard endpoints (CKD progression, stone events, final adult height, BMD) across health-economic settings.
 6. **Explore inner-ear-directed or gene-based strategies** (e.g., AAV delivery of *ATP6V0A4* to cochlea) using the knockout mouse/zebrafish models to address the currently irreversible hearing loss.
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Consensus Answer

Renal tubular acidosis, distal, 3, with or without sensorineural hearing loss (dRTA-3; OMIM #602722, MONDO:0011268) is a rare autosomal-recessive disorder caused by biallelic loss-of-function mutations in *ATP6V0A4* (7q34), encoding the $\alpha 4$ subunit of the apical vacuolar H^+ -ATPase of renal α -intercalated cells; loss of this proton pump abolishes distal urinary acidification, producing hyperchloremic normal-anion-gap metabolic acidosis with

hypokalemia, hypercalciuria/hypocitraturia, nephrocalcinosis, growth failure and rickets, plus variable (often later-onset) sensorineural hearing loss because the same pump acidifies inner-ear endolymph. It is diagnosed by inappropriately alkaline urine (pH >5.5) during acidosis (confirmed by acid-loading/furosemide testing) plus NGS gene-panel sequencing, and managed lifelong with oral alkali (bicarbonate/citrate, including prolonged-release ADV7103/Sibnaya), which corrects the biochemistry and improves growth, bone and quality of life but does not reverse established hearing loss.

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